

Holes and Vertical Asymptotes

A **vertical asymptote** is a line $x = a$ where the function grows without bound from at least one side. First simplify, but keep the original domain restrictions. Find the domain, hole, and vertical asymptote of

$$f(x) = \frac{2(x-1)}{(x-1)(x+2)}.$$



Approaching from the Left and Right

Use denominator signs to find the limits from each side of the vertical asymptote.

$$(a) \quad f(x) = \frac{1}{x-2} \qquad (b) \quad g(x) = -\frac{3}{(x+1)^2}.$$

Explain why the two sides have opposite directions in (a), but the same direction in (b).

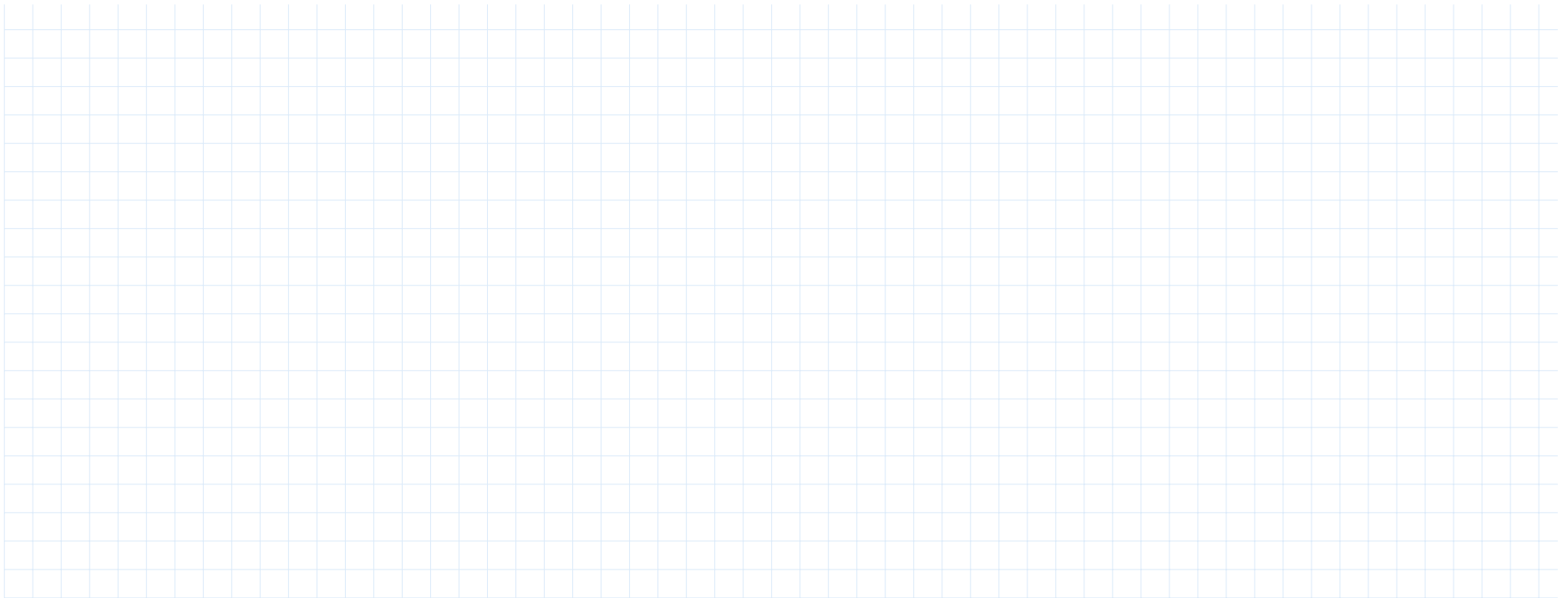


Horizontal Asymptotes

A **horizontal asymptote** $y = L$ describes $f(x) \rightarrow L$ as $x \rightarrow \infty$ or $x \rightarrow -\infty$. Compare numerator and denominator degrees: smaller gives $y = 0$; equal gives the leading-coefficient ratio; larger gives no horizontal asymptote.

$$(a) \frac{3x - 1}{x^2 + 4} \quad (b) \frac{6x^2 + 1}{2x^2 + 5} \quad (c) \frac{x^2 + 1}{x - 2}.$$

Find each horizontal asymptote, if any. State the limits at both ends in (a) and (b).



Combine Vertical and End Behavior

Find all vertical and horizontal asymptotes of

$$h(x) = \frac{2x - 1}{x + 3}.$$

Then find the limits as $x \rightarrow -3^-$, $x \rightarrow -3^+$, $x \rightarrow -\infty$, and $x \rightarrow \infty$.



Two Vertical Asymptotes

For

$$r(x) = \frac{x + 1}{(x - 2)(x + 3)},$$

find all asymptotes. Use factor signs to find the limit from each side of each vertical asymptote.



Can a Graph Cross an Asymptote?

A horizontal asymptote describes the ends of a graph, not a barrier.

$$q(x) = \frac{x}{x^2 + 1}.$$

Find the horizontal asymptote. Determine whether the graph crosses it, and sketch the graph near the crossing and at the ends.

